

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic

Project Execution & Progress Package

By

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and
Healthcare Support Using Data and AI

Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 5 Submission – Supporting Document

Table of Contents

1. Introduction
 2. Updated Project Plan
 3. Task Tracker
 4. Updated Timeline
 5. Progress / Status Report
 6. Updated Risk and Dependency Register
 7. Track Coordination Plan
 8. Summary of Major Project Issues Encountered
- References

1. Introduction

This Project Execution & Progress Package consolidates the Data Science track's project management artefacts following the completion of the updated Week 5 notebook (data preparation, feature engineering and baseline model development, extended with formal statistical testing and a documented cross-track exchange). It updates the Week 4 Data Science Track Project Plan and Week 4 Risk and Dependency Register, and this package's own earlier version, in light of what was actually delivered, and adds three new artefacts, a task tracker, a progress/status report, and a track coordination plan, to reflect the project's move from planning into active execution.

This package should be read alongside the Data Science Baseline Modelling Report, the Week 5 Project Summary and the Review of Week 4 Foundation; detailed technical findings are not repeated here.

2. Updated Project Plan

The Work Breakdown Structure defined in Week 4 is unchanged in scope, with one addition (2.5) reflecting work that emerged during execution but was not originally planned. Status has been added to each work package.

WBS ID	Work Package	Stage	Status
1.0–1.4	Problem Understanding (resource review, ML problem, feature candidates, Problem Definition Document)	Week 4	Complete
2.1	Resolve open data-quality questions (waiting_time_minutes)	Week 5	Complete
2.2	Conduct formal statistical and correlation analysis	Week 5	Complete
2.3	Complete feature encoding and scaling	Week 5	Complete
2.4	Build and evaluate a baseline logistic regression model	Week 5	Complete
2.5 (new)	Investigate and document the Sunday-appointment data-quality anomaly	Week 5	Complete (escalation pending)
3.0–3.4	Development – candidate ensemble models, tuning, fairness review	Week 6	Not Started
4.0–4.3	Testing and Refinement	Week 7	Not Started
5.0–5.3	Final Presentation	Week 8	Not Started

Work package 2.2 is now marked **Complete**: alongside the correlation analysis, five independent-samples t-tests were run against the target variable, formally validating the significance of the key numerical relationships. Chi-square testing of the categorical variables (gender, appointment_type, reminder_sent) was not part of this scope and is carried forward into Week 6.

3. Task Tracker

Task ID	Task	Week	Status
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T1	Review HealthConnect Appointment Dataset and Data Dictionary	4	Complete
T2	Define ML problem, target variable and candidate features	4	Complete
T3	Produce ML Problem Definition Document	4	Complete
T4	Produce Week 4 supporting registers and plan	4	Complete
T5	Resolve waiting_time_minutes leakage question	5	Complete
T6	Run data-quality and consistency checks (duplicates, ID uniqueness, date logic)	5	Complete
T7	Investigate Sunday-appointment anomaly and engineer is_sunday_appointment flag	5	Complete
T8	Engineer date-derived features (booking/appointment month, day-of-week, etc.)	5	Complete
T9	Implement patient-aware GroupShuffleSplit and confirm zero patient overlap	5	Complete
T10	Encode categorical features and scale numerical features (train-fit only)	5	Complete
T11	Train and evaluate logistic regression baseline	5	Complete
T12	Run formal statistical hypothesis tests (t-tests) on numerical predictors	5	Complete
T13	Exchange and cross-validate findings with the Data Analytics track	5	Complete
T14	Reconcile notebook documentation/README against executed results	5	Partially Complete
T15	Run chi-square tests of independence on categorical variables	6	Not Started
T16	Train and compare Random Forest and gradient boosting models	6	Not Started
T17	Implement GroupKFold cross-validation for hyperparameter tuning	6	Not Started
T18	Calibrate classification threshold with clinic stakeholders	6	Not Started

T19	Escalate Sunday- appointment anomaly to data owner / mentor	6	Not Started
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4. Updated Timeline

Week	Phase	Planned Deliverable	Status
Week 4	Problem Understanding	ML Problem Definition Document, registers, project plan	Delivered
Week 5	Analysis and Solution Design	Baseline model and analysis notebook	Delivered
Week 6	Development	Candidate model comparison report	Not Started
Week 7	Testing and Refinement	Final validated model and documentation	Not Started
Week 8	Final Presentation	Final presentation and portfolio update	Not Started

No milestone dates have changed. The additional Sunday-appointment investigation (WBS 2.5), the formal t-testing (T12) and the Data Analytics exchange (T13) were all absorbed within the existing Week 5 timebox and are not expected to affect the Week 6 start date; the outstanding documentation reconciliation (T14) and the chi-square testing of categorical variables (T15) carry forward as the first items of Week 6 work.

5. Progress / Status Report

Overall Data Science track status at the end of Week 5:

Work Area	Status	Comment
Data preparation	Green	Cleaning, missing-value treatment and leakage screening complete.
Feature engineering	Green	Date-derived features and the Sunday-appointment flag complete.
Train/test strategy	Green	Patient-aware split implemented and verified (zero patient overlap).
Baseline model	Green	Logistic regression trained and evaluated; results documented.
Statistical testing	Green	Correlation analysis and five t-tests complete for numerical predictors; chi-square tests for categorical variables carried into Week 6.
Documentation integrity	Amber	Reported metrics now match executed results throughout; the README's description of the train/test split method still needs correcting.
Cross-track coordination	Amber	Data Analytics exchange completed and findings cross-validated; Machine Learning Engineering, Project Management and Generative AI syncs still pending (see Section 7).
Stakeholder escalations	Red	Sunday-appointment anomaly not yet raised with the data owner or mentor.

Milestone M2 (“feature set finalised and baseline model trained”) is achieved. No item currently at Amber or Red status is assessed as a threat to the Week 6 start date, but the Red item (stakeholder escalation) should be actioned early in Week 6 given its dependency on an external party.

6. Updated Risk and Dependency Register

The Week 4 Risk and Dependency Register has been reviewed and updated below. Status changes and two newly identified risks (R7, R8) are shown; risk IDs are preserved from the Week 4 register for traceability.

6.1. Risk Register (Updated)

ID	Risk	Likelihood	Impact	Status (Week 4 → Week 5)
R1	waiting_time_minutes ambiguity / leakage risk	Medium	High	Open → Mitigated (excluded from feature set; a Week 5 t-test found no significant group difference, $p = 0.781$, supporting the decision; data-owner confirmation still pending)
R2	Cancellation-exclusion definition may not match clinic priorities	Low	Medium	Open (unchanged; not yet confirmed with stakeholders)
R3	Uneven model performance across demographic subgroups	Medium	High	Open (unchanged; fairness review deferred to Week 6)
R4	Synthetic dataset may not generalise	High	Medium	Accepted (unchanged)
R5	Technical debt from insufficient reproducibility discipline	Medium	Medium	Open – partially materialised (see R8), now improving as R8 is corrected; patient-aware, version-controlled pipeline continues to mitigate
R6	Cross-track schedule misalignment	Medium	Medium	Open → Partially Mitigated (Data Analytics sync completed this week; Machine Learning Engineering, Project Management and Generative AI syncs still pending)
R7	Sunday appointments (737 records) conflict with the Clinic Knowledge Base's stated opening hours	Confirmed	Medium	Open – mitigated with an engineered flag; escalation to data owner outstanding
R8	Notebook narrative/README reported metrics that do not match the	Confirmed	Medium	Open → Partially Resolved (reported metrics now match executed output)

	executed code's actual output			throughout the notebook; the README's description of the train/test split method remains uncorrected)
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6.2. Dependency Register (Updated)

ID	Dependency	Status (Week 4 → Week 5)
D1	ML Engineering depends on this track's target variable and feature set	Delivered (unchanged; feature set now finalised for the baseline, see Section 2)
D2	This track's modelling approach depends on data-owner clarification of waiting_time_minutes	Open → Superseded: a provisional decision (exclusion) was made without waiting for confirmation, given the Week 5 deadline; confirmation is still requested
D3	Interpretation of 'Cancelled' depends on the Clinic Knowledge Base (Generative AI track resource)	Satisfied (unchanged)
D4	This track's timeline depends on the Project Management track's WBS and High-Level Timeline	Pending → Delivered (Week 4 Project Plan; updated in this package, Sections 2 and 4)
D5	Future fairness review depends on continued availability of demographic fields	Satisfied (unchanged)

7. Track Coordination Plan

A first formal cross-track exchange, with Data Analytics, was completed during Week 5 (see Section 5); Machine Learning Engineering, Project Management and Generative AI syncs have not yet taken place. This section sets out a concrete coordination plan to complete that picture going into Week 6, rather than repeating the intention to coordinate without specifying how.

- **Machine Learning Engineering:** share the finalised feature list, the one-hot encoding scheme and the fitted-scaler approach (train-only fitting) from this package before Week 6 model development begins, so that system design work is not based on an outdated Week 4 assumption.
- **Project Management:** provide this package's updated Task Tracker and Timeline (Sections 3–4) for incorporation into the shared project timeline, and flag risk R7 (Sunday-appointment anomaly) as requiring a stakeholder decision.
- **Generative AI:** share the Sunday-appointment finding (R7), since it touches the same Clinic Knowledge Base that track relies on for describing clinic opening hours to users.
- **Data Analytics:** exchange completed this week – dashboard-derived no-show rates (by booking lead time, prior no-show history, distance and reminder status) were shared and found to corroborate this track's own t-test and correlation findings; continue this joint validation into the Week 6 model-comparison stage.
- **Proposed cadence:** a brief asynchronous written update (of the kind provided by this package and the Week 5 Project Summary) at the end of each week, supplemented by a short synchronous check-in before Week 6 model comparison work begins, given the cross-track dependencies (D1, D4) now converging at that point.

8. Summary of Major Project Issues Encountered

Issue	Description	Resolution / Status
Repeated-patient leakage risk	5,000 appointments across only 1,696 patients meant a standard random split risked patient overlap between train and test sets.	Resolved – patient-aware GroupShuffleSplit adopted; zero overlap confirmed
Sunday-appointment anomaly	737 appointments (14.7%) scheduled on Sundays, despite the Clinic Knowledge Base stating the clinic is closed that day.	Partially resolved – flagged as a feature; escalation to data owner outstanding (R7)
Documentation/code inconsistency	Notebook narrative and auto-generated README reported baseline metrics that did not match the actual executed code's output.	Substantially resolved – reported metrics now match executed output throughout the notebook; only the README's description of the split method remains uncorrected (T14, R8)
waiting_time_minutes ambiguity	Field description did not make clear whether values were available at booking time or only after the appointment.	Mitigated – excluded from the feature set as a precaution; formal confirmation from the data owner still requested (D2)

References

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